

番号	氏名
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課題

- [1] 玉の自由落下と雨粒の空気抵抗落下のグラフを手書きし、特徴を記せ.
- [2] 私のMacBook AirのCPU性能は何GFLOPSか？ 計算過程も記せ.
- [3] 卓上スパコンで何をするか、夢を語れ.

Any sufficiently developed technology is indistinguishable from magic.



Oxford Univ.,
Department of Materials
Sutton group(1992-3)

CRAY Y-MP2E/264
ベクトル型計算機
CPU数:2基
メモリ容量:512MB
HDD容量:62.7GB

CRAY T94/4
ベクトル型計算機
CPU数
メモリ容量:
HDD容量:100

n2000
 十算機
 5MHz
 128基
 28GB

HDD容量:1TB
OS:IRIX
理論演算性能:50 GFlops

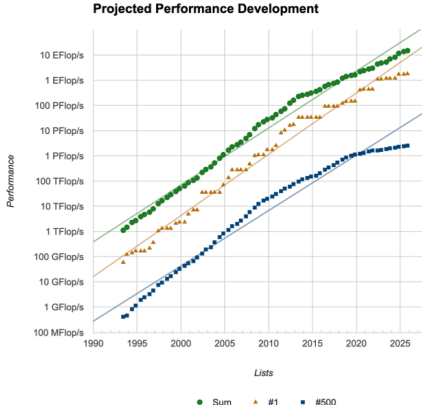
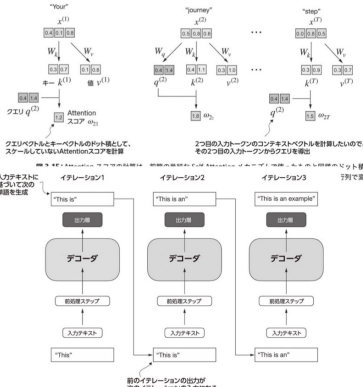
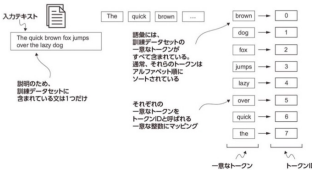


TP9400
大容量高速ストレージ
総記憶容量:21TB

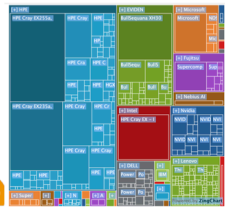
京大 化研

As simple as possible,
but not simpler.

top500 on Nov. 2025



Nov. 2020



Nov. 2025

LLM
Token

2015

2017

Attention is all you need.

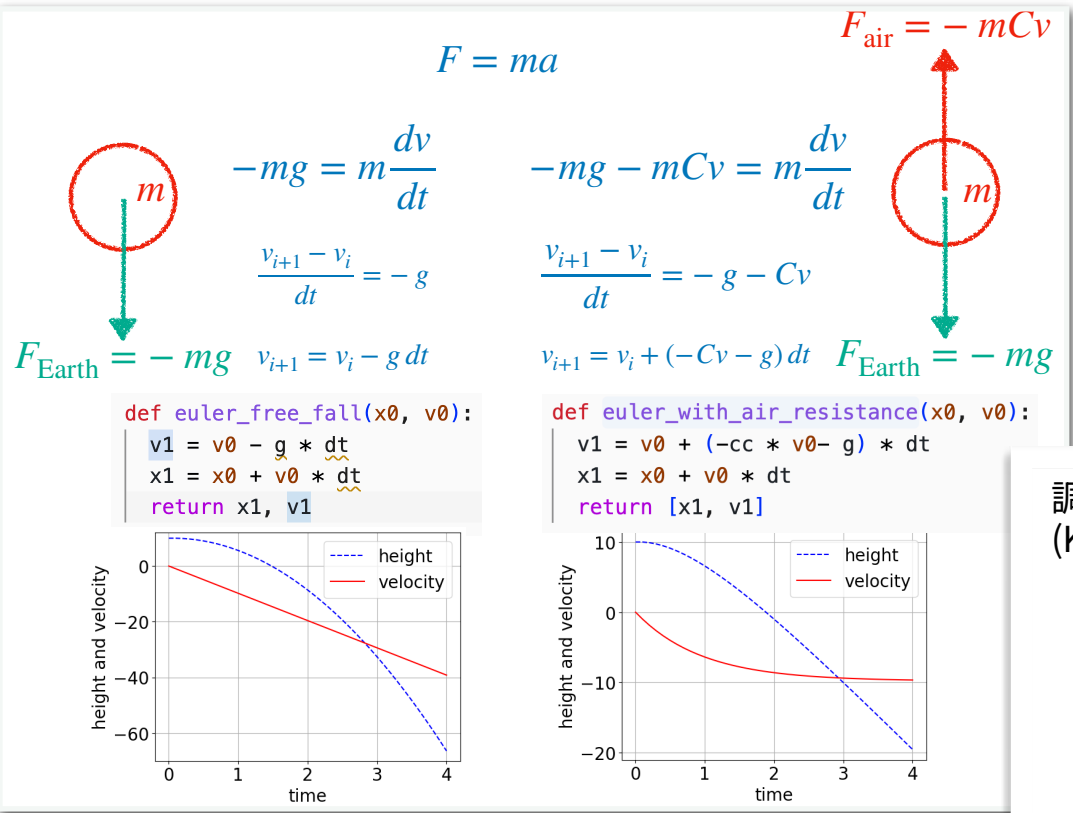
2022
Chat
G
P
T

<https://github.com/rasbt/LLMs-from-scratch>

重力のイメージ

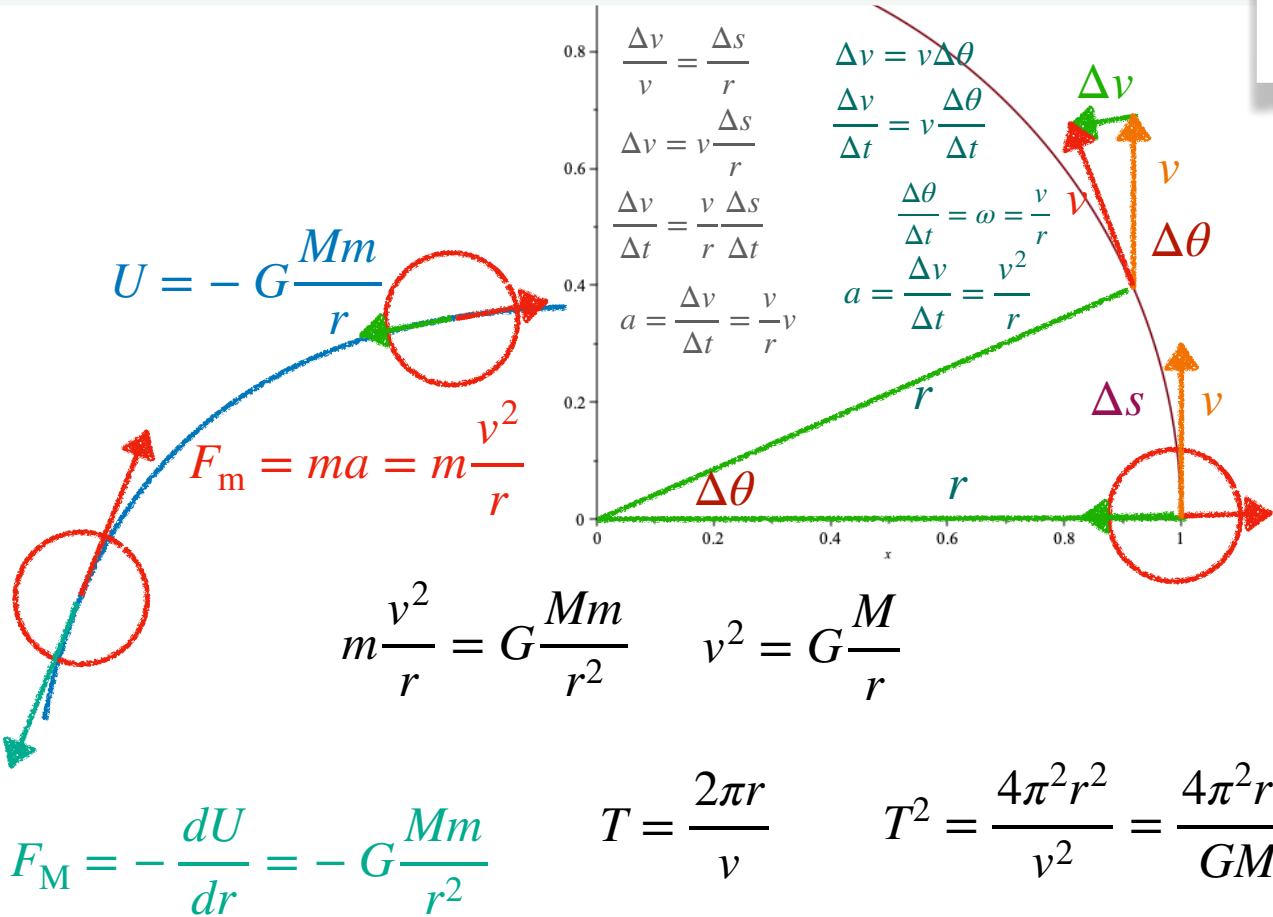
keyword: F=ma, free fall, air resistance,
Kepler, Harmonic law
source: kepler_mahmonic, SolarSystem.mw, d1_python_ode.ipynb

daddygongon 26/5/31



調和の法則 Law of Harmonies (Kepler's third law of planetary motion)

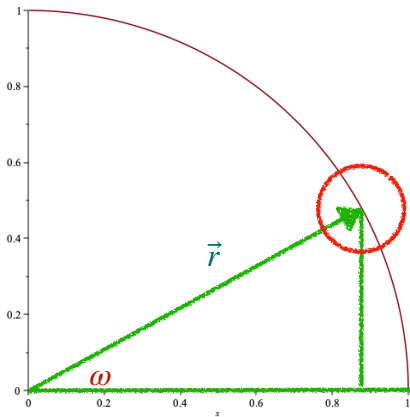
惑星	公転周期(T)	軌道長半径(r)	T^2/r^3 by Copilot	T^2/r^3 by you
水星	0.241	0.387	0.037	
金星	0.615	0.723	0.078	
地球	1.000	1.000	1.000	1.000
火星	1.881	1.524	0.296	
木星	11.86	5.203	1.000	
土星	29.46	9.537	0.930	
天王星	84.01	19.191	1.000	
海王星	164.8	30.07	1.000	



$$\vec{r} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} r \cos(\omega t) \\ r \sin(\omega t) \end{pmatrix}$$
$$\vec{v} = \frac{d\vec{r}}{dt} = \begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{pmatrix} = \begin{pmatrix} -r\omega \sin(\omega t) \\ r\omega \cos(\omega t) \end{pmatrix}$$
$$\vec{a} = \frac{d\vec{v}}{dt} = \begin{pmatrix} -r\omega^2 \cos(\omega t) \\ -r\omega^2 \sin(\omega t) \end{pmatrix}$$
$$\vec{a} = -\omega^2 \begin{pmatrix} r \cos(\omega t) \\ r \sin(\omega t) \end{pmatrix} = -\omega^2 \vec{r}$$

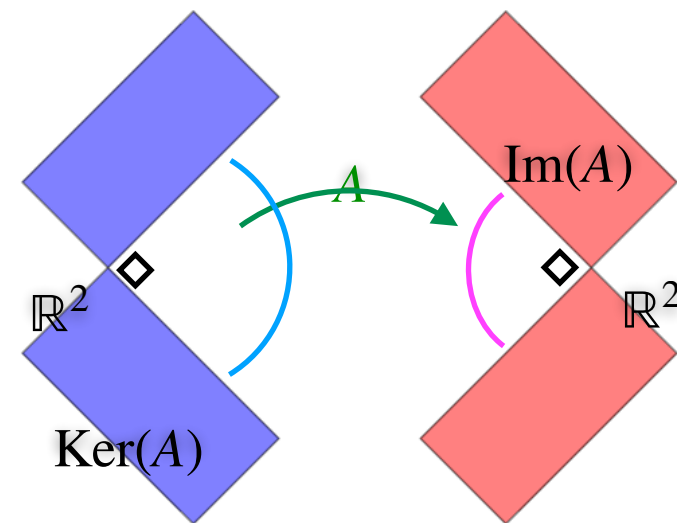
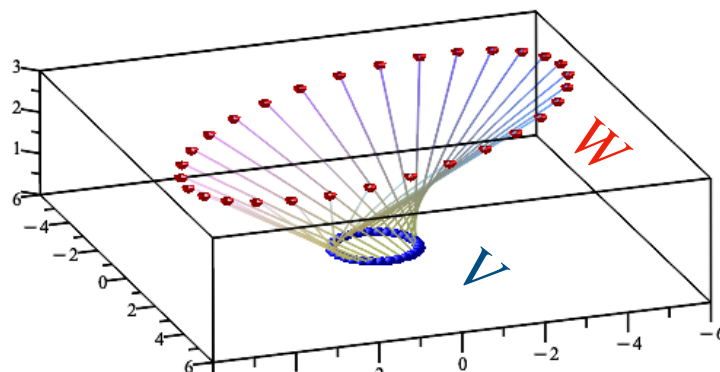
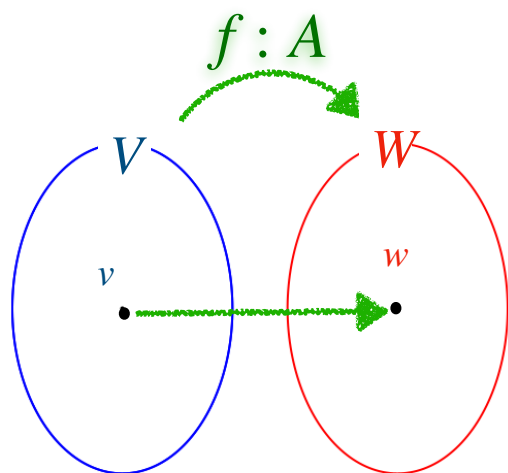
$$v = r\omega$$

$$a = r \left(\frac{v}{r} \right)^2 = \frac{v^2}{r}$$



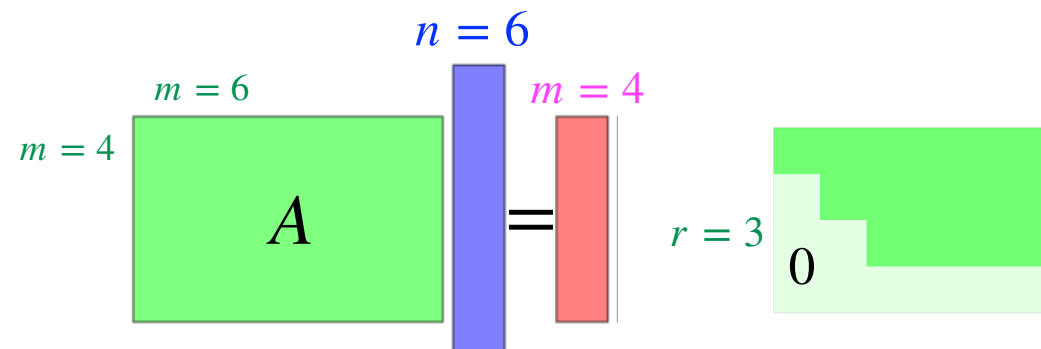
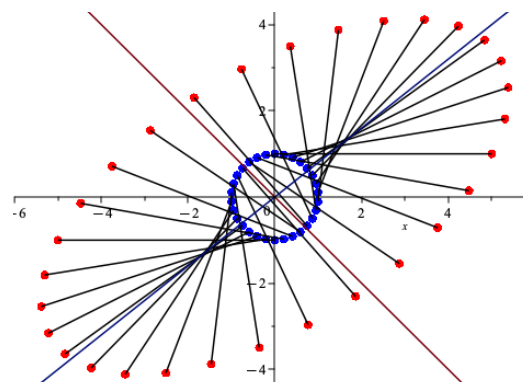
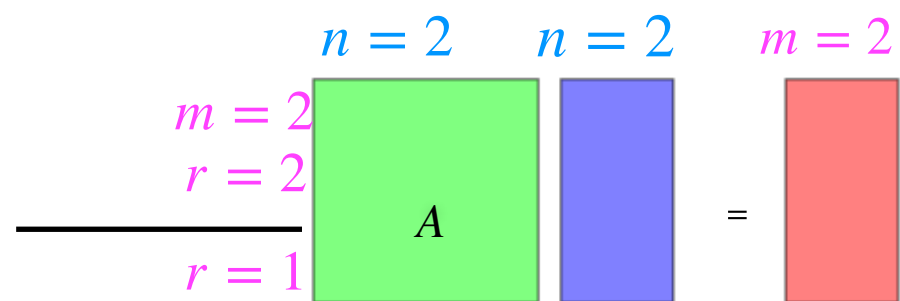
写像(Mapping)

$$\begin{pmatrix} 2 & 5 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} w_x \\ w_y \end{pmatrix}$$



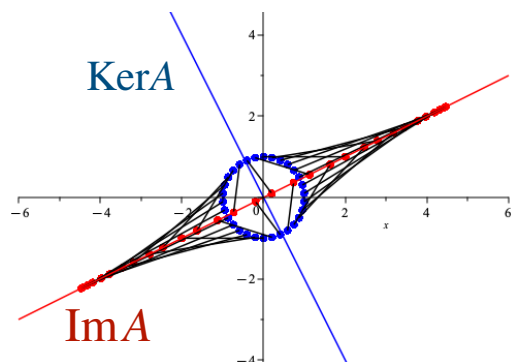
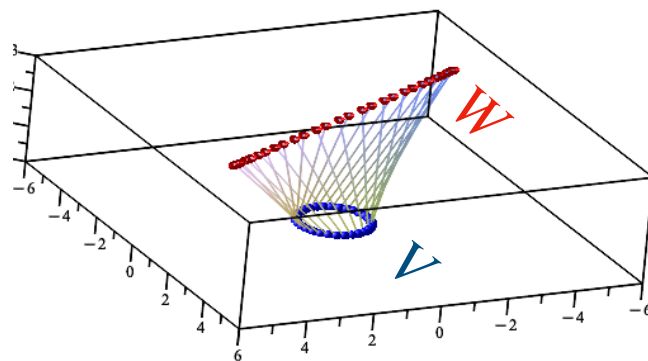
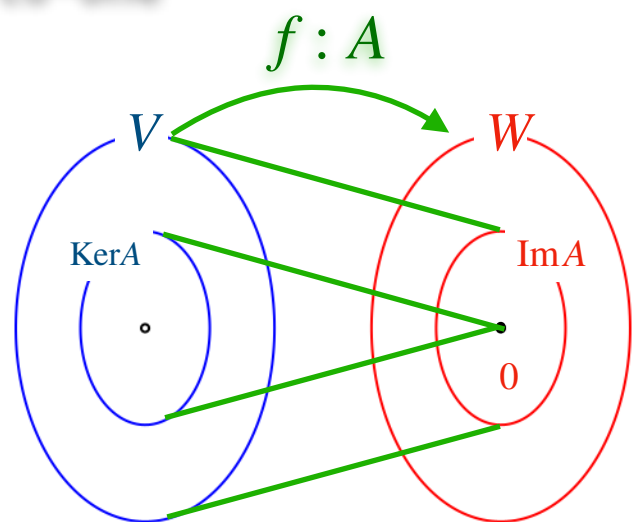
onto 全射

one-to-one 单射

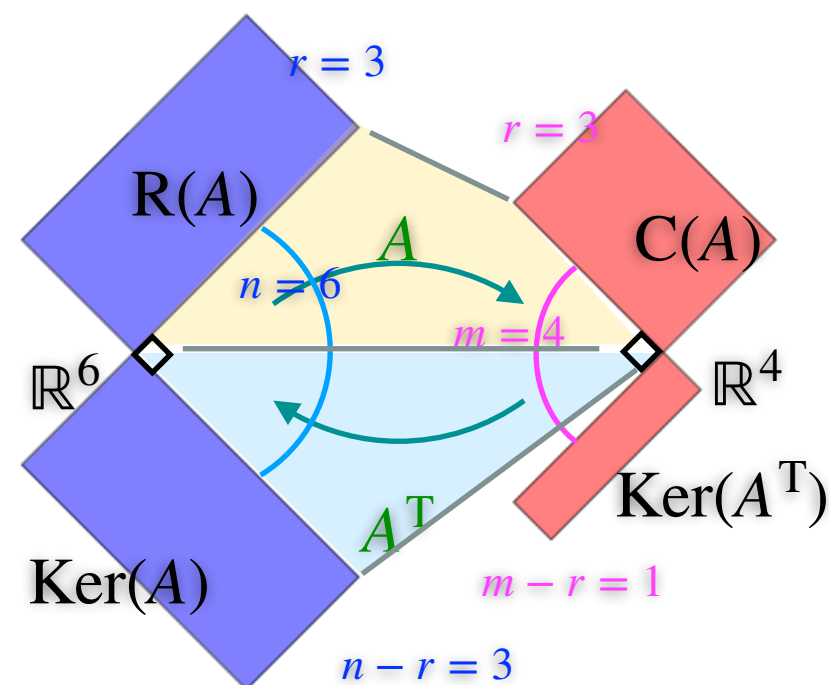


not onto

not one-to-one



$$\begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} w_x \\ w_y \end{pmatrix}$$



課題

- [1] ボールの自由落下と雨粒の空気抵抗落下のグラフを手書きし、特徴を記せ.
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• このMacBookAirのFLOPSは？

- 0.5 GFLOPS
- 5 GFLOPS
- 45 GFLOPS

MacBook Air(M4,2025)

```
# at Apple M4
> clang lapack.c -framework Accelerate
> ./a.out
1000
n=1000 [dim] 0.015 [sec]
```

FLOPS, フロップス,
FLoating-point OPerations / Second
1秒間に浮動小数点数演算が何回できるか

```
35 start = clock();
36 dgesv_(&n, &nrhs, a, &lda, ipiv, b, &ldb, &info);
37 end = clock();
38 printf("%5d [dim] %10.4f [sec] #LAPACK\n",
39        (int)n, (double)(end-start)/CLOCKS_PER_SEC);
```

for-loops for Matrix Inverse

```
for i from 1 to n do #i行目
  T[i]:=Matrix(array(1..n,1..n,identity)):
  #i番目の消去行列
  for j from i+1 to n do
    am:=A[j,i]/A[i,i]; #i行の要素から, i+1
    #行目の先頭を消す係数
    T[i][j,i]:=-am; #i番目の消去行列に
    #要素を入れる
    L[j,i]:=am; #LTMの要素
    for k from 1 to n do
      A[j,k]:=A[j,k]-am*A[i,k]; #もとの行列を
      #UTMに
    end do;
  end do;
end do;
```

3重for-loop

Vender and type	cpu	Time [sec]	Speed [GFLOPS]	Time x Speed
Convex C-4640	1	0.948	0.705	
Convex C-4640	4	0.278	2.404	
Cray C90	1	0.775	0.863	
Cray C90	16	0.092	7.270	
DEC 3000 Alpha	1	7.656	0.087	
IBM Power2	1	3.970	0.168	
IBM RISC sys/600	1	11.840	0.056	
SGI power challenge	1	3.371	0.198	
Apple MacBookAir(M1,2020)	2	0.056	???	
Apple MacBookAir(M2,2025)		0.031	???	
Apple MacBookAir(M4,2025)			???	

